

Internet of Things (IoT) -- Guided Notes ANSWER KEY

Unit tp-2.12 | CompTIA Network+ N10-009 Obj. FC0-U71 2.12 | N10-008 FC0-U71 2.12

For instructor use / distribute after students complete the notes.

Question 1

IoT stands for _____ of _____. It refers to everyday physical devices -- beyond traditional computers -- that are connected to the _____ and can send or receive data. Examples include smart _____, wearable _____, and industrial _____ sensors.

Answer: Internet of Things; Internet; thermostats; fitness trackers; equipment.

Question 2

IoT devices commonly communicate using low-power wireless protocols such as _____ (personal area network), _____ (short-range wireless), _____ (long-range low-power), and Wi-Fi. The choice of protocol depends on required _____, power constraints, and _____ range.

Answer: Zigbee; Bluetooth (BLE); LoRa / LoRaWAN; bandwidth; communication.

Question 3

IoT devices often run _____ operating systems with minimal resources (small CPU, limited RAM). Because they are difficult to _____, many become security liabilities as vendors stop releasing _____ patches. This is known as the IoT _____ problem.

Answer: Embedded; update (patch); firmware; security (attack surface).

Question 4

An IoT _____ is a hub device that collects data from local IoT sensors and aggregates or preprocesses it before sending to the cloud. This reduces _____ and allows some functions to work even without an active _____ connection. The gateway also acts as a _____ between IoT protocols and IP networking.

Answer: Gateway; bandwidth; Internet; bridge (translator).

Question 5

Edge computing in IoT means processing data _____ to where it is generated rather than sending all data to a remote cloud. This reduces _____, saves _____ bandwidth, and allows faster response. A self-driving car is an example -- it cannot wait _____ ms for the cloud to decide whether to brake.

Answer: Closer (at the device or local gateway); latency; network; 100+.

Question 6

The attack surface of IoT devices is large because they often have: default _____ that users never change, no _____ update mechanism, exposure on the _____ if firewalls are not configured, and minimal _____ logging for anomaly detection.

Answer: Passwords; automatic patch/firmware; Internet; security.

Question 7

A _____ attack leverages thousands of compromised IoT devices (botnet) to flood a target server with traffic, overwhelming it. The 2016 _____ botnet used IoT cameras and routers with default credentials to launch massive DDoS attacks. Organizations can mitigate this by segmenting IoT devices on a separate _____ VLAN.

Answer: DDoS; Mirai; isolated (dedicated).

Question 8 -- Real World Application

REAL WORLD: A hospital is planning to deploy 200 internet-connected patient monitoring sensors throughout patient rooms. The IT security team is concerned about the security risks. Identify three specific security risks introduced by these IoT devices and provide a corresponding mitigation strategy for each.

Answer: Risk 1 -- Default credentials: many IoT devices ship with default admin passwords. Mitigation: change all default credentials to unique strong passwords before deployment; document them in a secure password manager. Risk 2 -- Unpatched firmware: medical IoT vendors may stop releasing firmware updates, leaving known vulnerabilities unpatched. Mitigation: establish a vendor contract requiring firmware support lifecycle; monitor CVE databases for the device models; isolate on a VLAN so an exploited device cannot reach patient records on the main network. Risk 3 -- Network exposure / lateral movement: a compromised sensor on the hospital network could be used as a pivot point to attack EMR servers. Mitigation: segment all IoT devices on an isolated VLAN with firewall rules permitting only the specific traffic needed (data upload to the monitoring server), blocking all other lateral traffic.